

of this rod, we choose an infinitesimally small mass dm of elemental length dx at a distance x from the origin.

λ is the linear mass density (i.e. mass per unit length) of the rod. $\lambda = \frac{M}{\ell}$

The mass of small element (dm) is, $dm = \frac{M}{\ell} dx$

Now, we can write the centre of mass equation for this mass distribution as,

$$x_{CM} = \frac{\int x dm}{\int dm}$$

$$x_{CM} = \frac{\int_0^{\ell} x\left(\frac{M}{\ell} dx\right)}{M} = \frac{1}{\ell}\int_0^{\ell} x dx$$

$$= \frac{1}{\ell}\left[\frac{x^2}{2}\right]_0^{l} = \frac{1}{\ell}\left(\frac{\ell^2}{2}\right)$$

$$x_{CM} = \frac{\ell}{2}$$

As the position $\frac{\ell}{2}$ is the geometric centre of the rod, it is concluded that the centre of mass of the uniform rod is located at its geometric centre itself.

5.1.6 Motion of Centre of Mass

When a rigid body moves, its centre of mass will also move along with the body. For kinematic quantities like velocity (v_{CM}) and acceleration (a_{CM}) of the centre of mass, we can differentiate the expression for position of centre of mass with respect to time once and twice respectively. For simplicity, let us take the motion along X direction only.

$$\vec{v}_{CM} = \frac{d\vec{x}_{CM}}{dt} = \frac{\sum m_i\left(\frac{d\vec{x}_i}{dt}\right)}{\sum m_i}$$

$$\vec{v}_{CM} = \frac{\sum m_i \vec{v}_i}{\sum m_i} \qquad (5.7)$$

$$\vec{a}_{CM} = \frac{d}{dt}\left(\frac{d\vec{x}_{CM}}{dt}\right) = \left(\frac{d\vec{v}_{CM}}{dt}\right) = \frac{\sum m_i\left(\frac{d\vec{v}_i}{dt}\right)}{\sum m_i}$$

$$\vec{a}_{CM} = \frac{\sum m_i \vec{a}_i}{\sum m_i} \qquad (5.8)$$

In the absence of external force, i.e. $\vec{F}_{ext} = 0$, the individual rigid bodies of a system can move or shift only due to the internal forces. This will not affect the position of the centre of mass. This means that the centre of mass will be in a state of rest or uniform motion. Hence, $\vec{v}_{CM}$ will be zero when centre of mass is at rest and constant when centre of mass has uniform motion $\left(\vec{v}_{CM} = 0 \text{ or } \vec{v}_{CM} = \text{constant}\right)$. There will be no acceleration of centre of mass, $\left(\vec{a}_{CM} = 0\right)$.

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